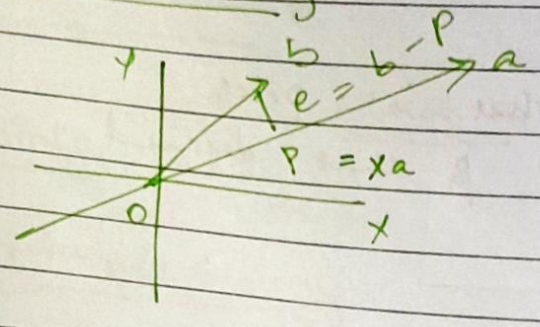
Machine Learning.

Linear Algebra



Maximal Margin Classifier.

$$Ax = b$$

$$a^T e = 0 \qquad P = xa$$

$$a^T(b - xa) = 0$$

$$x\, a^T a = a^T b$$

$$x = \frac{a^T b}{a^T a} \text{ — Matrix} \qquad a^T a = \text{Scalar.}$$

$$P = a\left(\frac{a^T b}{a^T a}\right)$$

$$P = PMb$$

$$PM = a\left(\frac{a^T}{a^T a}\right)$$

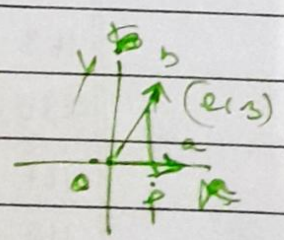
Projection Matrix in 1D

Example

$$b = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

Project b onto a

$$a = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$$



$$P =$$

$$x = \frac{a^T b}{a^T a}$$

$$a^T = [4\ 0]$$

$4 \quad 0 \qquad 2 \quad 3$

$a^T b = \begin{bmatrix} 4 & 0 \end{bmatrix} \begin{bmatrix} 2 & 3 \end{bmatrix}$ $\quad 2\times1 \quad 1\times2$

2×2

$= 8+12$

$\begin{bmatrix} 8+12 & 0 \end{bmatrix}$

$= \begin{bmatrix} 20 & 0 \end{bmatrix}$

$a^T a = \begin{bmatrix} 4 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 0 \end{bmatrix}$

$16+0$

$\rightarrow \quad \downarrow$

$4 \quad 0 \qquad 2$

$\qquad\qquad 3$

$8+12$

20

$\begin{bmatrix} 8 & 0 \\ 12 & 0 \end{bmatrix}$ 2×2

$a^T b$

$P_m = a\left(\frac{a^T}{a^T a}\right)$

$= \begin{bmatrix} 4 \\ 0 \end{bmatrix} \left(\frac{\begin{bmatrix} 4 & 0 \end{bmatrix}}{16} \right)$

$P = a\left(\frac{a^T b}{a^T a}\right)$

$= \begin{bmatrix} 4 \\ 0 \end{bmatrix} \frac{\begin{bmatrix} 8 & 0 \\ 12 & 0 \end{bmatrix}}{16}$

$=$

$$\frac{a^T}{a^T a} = \frac{\begin{bmatrix} 4 & 0 \end{bmatrix}}{16}$$

$$= \begin{bmatrix} \frac{1}{4} & 0 \end{bmatrix}$$

$$a \begin{bmatrix} \frac{1}{4} & 0 \end{bmatrix}$$

$$\frac{1}{4} \quad \frac{1}{4} \quad 0$$

$$0$$

$$\begin{bmatrix} 4 \\ 0 \end{bmatrix} \begin{bmatrix} \frac{1}{4} & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$PM = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$p = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$2 \times 2 \qquad 1 \times 2$

$$\rightarrow \begin{bmatrix} 5 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 \times 2 + 0 \times 2 \\ 3 \times 0 + 0 \times 3 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \\ 0 \end{bmatrix} \checkmark$$